

--- Inclusion Criteria ---

- Adults aged 18 years and older are eligible.
- Women of childbearing potential must have a negative pregnancy test at screening and before vaccination.
- Acceptable contraception methods include highly effective hormonal contraception, intrauterine devices, sterilization, or complete abstinence (only if this is the subject's normal and preferred lifestyle).
- Pregnancy tests are conducted at screening, pre-dose 1, and pre-dose 2.

--- Exclusion Criteria ---

- Prior receipt of any COVID-19 vaccine or investigational product for COVID-19.
- History of SARS-CoV-2 infection, confirmed by RT-PCR or serology.
- Participation in another clinical trial within the last 28 days, or receipt of another investigational vaccine within 12 weeks.
- Use of immunosuppressive therapy, cytotoxic drugs, or systemic corticosteroids within 6 months prior to enrollment.
- Clinically significant history of HIV, HBV, or HCV infection, uncontrolled cancers, significant cardiovascular, hepatic, or renal disease.
- Psychiatric exclusions: uncontrolled schizophrenia, bipolar disorder, major depressive episode with suicidal ideation, or other severe psychiatric conditions that could interfere with study participation.
- History of immune-mediated disease such as autoimmune thyroiditis, multiple sclerosis, systemic lupus erythematosus, or rheumatoid arthritis (unless mild and stable).
- Receipt of monoclonal antibodies or convalescent plasma for COVID-19 treatment within 3 months.

--- Primary Endpoints ---

- Vaccine efficacy against first occurrence of virologically confirmed COVID-19 of any severity, starting 14 days after dose 2.
- Solicited local and systemic adverse events are collected for 7 days post-vaccination.
- Unsolicited adverse events are collected for 28 days post-vaccination.
- Medically attended adverse events are collected through 6 months after dose 2.
- Serious adverse events (SAEs) and adverse events of special interest (AESIs) are collected for up to 12 months after dose 2.

- Investigators must report SAEs to the sponsor within 24 hours of becoming aware of the event.

--- Secondary Endpoints ---

- Prevention of moderate-to-severe COVID-19 occurring at least 14 days after dose 2.
- Prevention of asymptomatic SARS-CoV-2 infection confirmed by PCR or serology.
- Burden of disease endpoint, combining incidence and severity of COVID-19.
- Vaccine efficacy stratified by age groups, particularly ≥ 65 years.
- Immunogenicity endpoints include neutralizing antibody titers, binding antibodies, and T-cell responses measured by ELISpot/ICS assays.
- Monitoring of vaccine efficacy against emerging variants, including but not limited to Alpha (UK strain), Beta, and Delta variants.
- Evaluation of durability of immune responses at 6 and 12 months post-vaccination.

--- Vaccine Delay ---

- Vaccination should be postponed in subjects with acute illness, fever $\geq 38^{\circ}\text{C}$, or significant clinical worsening of underlying disease.
- Mild illness (e.g., low-grade fever, mild upper respiratory infection) does not necessarily require delay and is at the discretion of the investigator.
- If a subject develops COVID-19 after the first dose, the second dose should be deferred until at least 2 weeks after recovery and discontinuation of isolation.
- Other vaccines should not be given within 14 days prior to or after dose 1, and within 28 days before or after dose 2, except in urgent cases.

--- Treatment Arms ---

- Randomized, observer-blinded, placebo-controlled trial.
- Two treatment arms: (1) CVnCoV 12 μg mRNA vaccine, (2) placebo (saline solution).
- Stratified randomization by age groups (18–60 years and ≥ 61 years) to ensure balanced distribution.
- Trial includes Phase 2b ($n=4,000$) and Phase 3 ($n=32,500$) cohorts across multiple countries.
- Blinding procedures ensure that investigators, site staff, and participants remain blinded to treatment assignment, except for unblinded pharmacists preparing the doses.

--- Dosing Schedule ---

- Participants receive 2 intramuscular injections into the deltoid muscle: Day 1 and Day 29.
- Each active dose contains 12 µg of CVnCoV mRNA vaccine in lipid nanoparticles.
- Placebo group receives 0.9% sodium chloride solution (saline).
- Vaccine must be stored frozen at -80°C and thawed before administration.
- Once thawed, the vaccine is stable at 2–8°C for up to 24 hours and must not be refrozen.
- Preparation must be done under aseptic conditions by trained pharmacists; unused vials must be returned for accountability and disposal.

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